An Approach for Efficient Particle Picking in Cryo-ET Tomograms

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Period Covered: October 2025 - June 2026

Abstract

Cryo-electron tomography (Cryo-ET) allows the visualization of macromolecular structures in their native cellular environments at near-atomic resolution. However, particle picking—the identification and extraction of individual macromolecules from noisy tomograms—remains a challenging task. This report presents a detailed examination of Cryo-ET, with an emphasis on evaluating different strategies for particle picking in tomographic datasets. The study compares these approaches in terms of efficiency, accuracy, and scalability for automated, large-scale structural analysis. Finally, it discusses the current strengths and limitations of existing methods and outlines prospective directions for improving performance and reliability in future developments.

1 Introduction

Understanding the three-dimensional structure of macromolecules such as proteins, nucleic acids, and large molecular complexes is fundamental to explaining how biological systems operate at the molecular level. Structure reveals mechanism: it explains how enzymes catalyze reactions, how receptors recognize ligands, and how ribosomes translate genetic information into proteins.

In recent years, the development of advanced imaging and computational methods has made it possible to explore biological structures at near-atomic resolution. Among these, cryo-electron microscopy (cryo-EM) has emerged as a cornerstone technique in structural biology. By preserving samples at cryogenic temperatures in a vitreous layer of ice, cryo-EM enables imaging of macromolecules in a state that closely resembles their natural environment (without the need for crystallization or staining).

Despite its transformative impact, cryo-EM still faces significant computational and technical challenges. The inherently low signal-to-noise ratio (SNR) of cryo-EM images complicates the extraction of structural information, requiring advanced algorithms and vast numbers of particles to achieve high-quality reconstructions. This difficulty motivates the exploration of synthetic data generation and machine learning strategies to improve the reconstruction pipeline and particle identification processes.

This report explores the computational approaches underpinning the generation of synthetic 3D cryo-EM data, their role in improving image reconstruction and particle

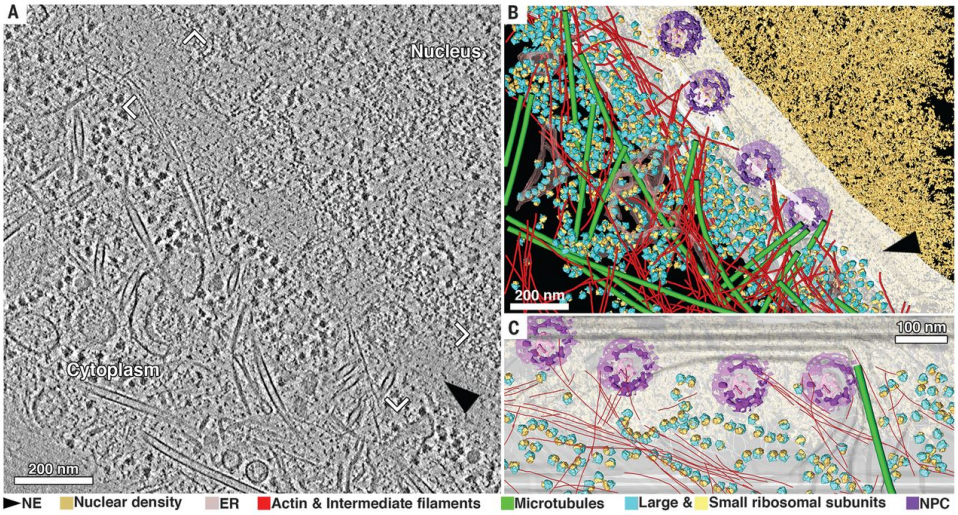

Figure 1: The nuclear periphery of a HeLa cell revealed by cryo-ET. (A) Tomographic slice. (B) Annotated view of the tomographic data. (C) Cross-section view of the segmentation. (Mahamid et al., 2016).

picking, and the challenges that remain. The next section provides the scientific background necessary to understand the principles, limitations, and ongoing developments in this field.

2 Scientific Background

2.1 Principles of Cryo-Electron Microscopy

Cryo-electron microscopy (cryo-EM) is a structural imaging technique that captures biological macromolecules preserved in a thin layer of vitrified ice at liquid nitrogen temperatures. Because the sample is maintained in a non-crystalline, hydrated state, cryo-EM avoids the artifacts introduced by traditional fixation or staining methods, providing images that represent the molecule's near-native conformation.

In single-particle cryo-EM, thousands to millions of two-dimensional (2D) projection images are recorded, each depicting the same type of macromolecule in a random orientation. These images are computationally aligned and averaged to reconstruct a three-dimensional (3D) electron density map. Advances in electron detectors, noise reduction, and image-processing algorithms, particularly those based on Bayesian inference and deep learning, have made it possible to reach near-atomic resolutions in favorable cases.

2.2 Cryo-Electron Tomography (Cryo-ET)

While single-particle analysis provides detailed information about isolated molecular structures, cryo-electron tomography (cryo-ET) extends the method to capture complex biological systems *in situ*. In cryo-ET, a single specimen is imaged from multiple tilt angles, and these projections are combined to generate a 3D reconstruction known as a tomogram.

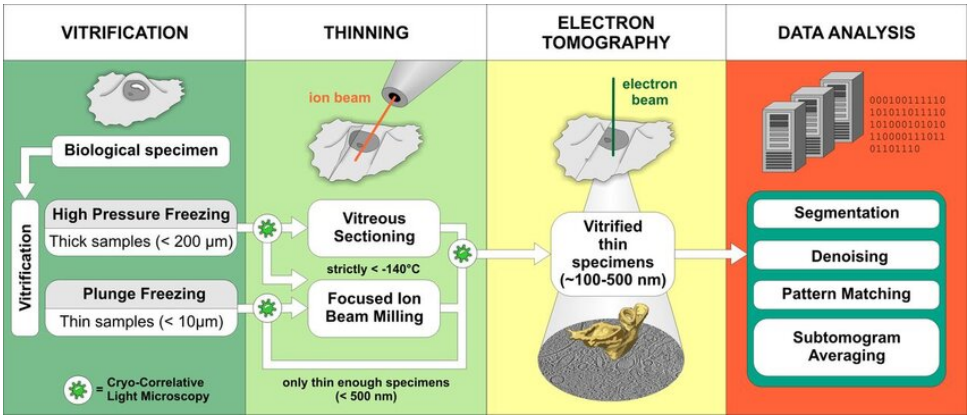

Figure 2: Schematic representation of the cryo-ET workflow (Lučić et al., 2013).

Unlike single-particle cryo-EM, cryo-ET visualizes molecular assemblies directly within their cellular context, preserving spatial relationships among organelles, membranes, and protein complexes. This makes cryo-ET particularly powerful for studying subcellular organization, viral infection mechanisms, and molecular interactions within tissues. However, this contextual advantage comes with a trade-off: lower resolution and image contrast due to limitations in electron dose and sample thickness.

Cryo-ET tilt series are collected under strict dose limits to avoid radiation damage, yielding tomograms with very low SNR.

2.3 Computational Challenges in Cryo-EM

Despite major advances in instrumentation, the analysis of cryo-EM data remains computationally demanding. One of the central problems is the low signal-to-noise ratio (SNR) inherent to electron microscopy images, as biological materials scatter electrons weakly. The resulting micrographs often appear extremely noisy, obscuring particle boundaries and making it difficult to distinguish true features from background.

This challenge directly impacts key steps in the cryo-EM processing workflow, particularly particle picking, the identification of individual macromolecules within a micrograph. Accurate particle picking is essential for high-quality 3D reconstruction, yet traditional correlation-based methods (such as template matching) are slow, laborious, and prone to bias. Moreover, the lack of labeled data for training machine learning algorithms exacerbates these difficulties.

To overcome these limitations, researchers are increasingly turning to synthetic data generation, a computational approach that simulates cryo-EM images with known ground truth positions and orientations. These synthetic datasets allow for controlled algorithm evaluation and provide valuable training data for deep learning models aimed at automating particle picking and improving 3D reconstructions. The next section will explore this process of synthetic image generation in depth, outlining its objectives, methodologies, and recent advancements.

2.4 Particle Picking in tomograms

As commented, Cryo-ET produces 3D volumes (tomograms) of specimens at nanometer resolutions. An essential step in cryo-ET analysis is particle picking: locating coordinates

of identical macromolecular complexes within the tomogram for further processing (like subtomogram averaging).

In SPA (single-particle analysis) this task is significantly less challenging, where particles are typically isolated and well-separated on a thin ice layer, making them easier to detect in 2D micrographs. In cryo-ET, the particles reside in a crowded cellular volume with many overlapping structures and heterogeneous molecular species. So, identifying a specific protein amidst the dense cellular milieu is non-trivial. In addition, the characteristic noise produced by this method can mask smaller particles, making manual or automated detection error-prone. Due to this problem, most of the recent methods are centered on improving the resolution of the input to go deep in tiny particles.

Researchers have developed several strategies to tackle particle localization in tomograms. This can be categorized into four main approaches: manual picking, template matching, semi-automated workflows and deep learning-based methods. Each approach has distinct advantages and limitations, and in practice they are often used in combination.

2.4.1 Manual particle picking (human-driven identification)

This is the most straightforward approach, where an expert visually inspects the tomogram and marks each particle's location by hand. Manual picking can be aided by prior knowledge of what the particle looks like (e.g. a ∼20 nm ribosome blob) or where it might be (e.g. along a membrane for a membrane protein complex).

As advantages, this method can exclude false signals when done carefully. For very recognizable large structures, a trained person can identify many true particles that automated methods might miss. By contrast, manual picking is tedious, slow and not scalable. A single tomogram (which might be $\sim 1k^3$ voxels) can take hours to comb through. In large studies with dozens or hundreds of tomograms, manual picking becomes the bottleneck, often taking months if attempted exhaustively.

Indeed, manual picking in cryo-ET is so challenging that it is usually reserved for providing a small initial training set or validating automated results, rather than picking every particle.

2.4.2 Template Matching (automated cross-correlation search)

Template matching is a classic computational approach to automated picking. The idea is to use a known 3D template of the particle (for example, a previously determined structure or a preliminary subtomogram average) and scan the tomogram for locations that resemble this template. This is typically done by sliding the template through the volume and computing a cross-correlation score at every position (and potentially every orientation). High correlation peaks indicate regions where the structure of the tomogram locally matches the template, i.e. likely particle locations.

More precisely, the tomogram $T(x, y, z)$ and a template density $M(x, y, z)$ (often low-pass filtered to tomogram resolution) are used to compute a correlation volume $C(x, y, z)$, where e.g. $C(x, y, z) = (T * M)(u, v, w)$ measures how well the template fits at location (u, v, w). Efficient implementations use Fast Fourier Transforms (FFTs) to perform 3D convolution, and sometimes multi-scale approaches to first find coarse matches and then refine them. Because the particle can have any orientation in the tomogram, one approach is to rotate the template in many orientations and perform the search for each, or equivalently to rotate the tomogram or use tricks like spherical harmonic correlations.

This is computationally heavy; searching a 3D volume in all orientations is orders of magnitude more work than 2D template matching in SPA. However, it is a well-defined brute-force strategy that does not require prior training data, only a template.

As advantages, template matching is fully automated and does not require any manual annotation of particles in the tomogram. If a reliable template is available, it can be run on large batches of tomograms in a high-throughput manner. It will detect particles even in very noisy data as long as the signal, when averaged over the whole particle volume, is above noise. It's especially useful when the target complex's structure is known (e.g., looking for ribosomes, one can use a ribosome density map as template).

By contrast, this method often produces many false positives and requires careful post-processing. In a crowded cell, many locations may spuriously correlate with the template due to random noise or similar-shaped structures, yielding high scores even though they are not the target particle. One must choose a correlation score threshold that balances sensitivity and precision, which is tricky. Often a large excess of candidates is picked and then pruned by other criteria (such as distance from membranes if the protein should only be at membranes, or by checking subtomogram average quality).

Despite these issues, template matching remains widely used, especially in projects where throughput is critical or deep learning methods are not feasible to train. For example, in a high-throughput pipeline dealing with many similar tomograms (e.g., a large set of lamellae where the goal is to find ribosomes in all of them), template matching with a known ribosome map is a practical solution. It provides a baseline set of coordinates that can be refined. However, as we'll see, newer methods aim to outperform template matching in both speed and accuracy.

2.4.3 Semi-Automated and iterative workflows

Semi-automated approaches are between fully manual and fully automatic methods. These typically involve an initial manual step or user guidance, followed by an automated routine to propagate that information and pick more particles. The purpose is to significantly reduce manual labor while avoiding the need to provide external template or large training set a priori.

One common method is to follow the strategy "pick a few, find many", where a user manually picks a small number of particles (in one or a few tomograms) as examples of the particle of interest. These picked subvolumes are then used in one of two ways:

- **As a template.** The user's manual picks can be aligned and averaged to create an initial method (a rough estimate of the particle density). This template is then used for standard template matching across dataset. Essentially, the method bootstraps itself: a few manual picks yield and average that allows automated cross-correlation search for more.

- **As training data.** The manual picks are used to train a classifier that can distinguish particles from background. This classifier is then run on the whole tomogram to detect particles automatically, and the results can be refined in further rounds. This approach has become more prominent with the rise of machine learning.

Another form of semi-automation is interactive segmentation or thresholding: the user might adjust a density threshold to segment out high-density regions in the tomogram (where particles likely reside) and then automatically detect connected components as

candidate particles. The user then verifies or cleans this list. This is not very specific but can be a quick way to get initial candidates for large, high-contrast particles (e.g., segment out all virus-sized objects).

Overall, semi-automated picking serves as a pragmatic middle ground, employed in many practical tomography studies: the user gets the process started and ensures quality control, while the algorithm does the bulk of the picking. This approach was common before the advent of large-scale deep learning and is still useful when one cannot fully rely on a generic pre-trained model.

2.4.4 Deep Learning-Based automated picking

In recent years, deep learning has revolutionized particle picking in cryo-EM and is making a similar impact in cryo-ET. These methods use neural networks to learn features of the particles directly from data, often outperforming template matching in both detection accuracy and speed. There are several classes of deep learning approaches for cryo-ET particle picking:

- **Supervised Deep Learning (Fully Annotation-Based).** These methods require a training set of tomograms where particle locations are labeled. A neural network is trained to predict those locations on new data. Usually, the network architecture is either a semantic segmentation model (like a 3D U-Net) that produces a probability map for particle presence, or an object detection model that outputs coordinates (and possibly sizes or classifications) of particles. An example of this method is DeepFinder, that uses a 3D U-Net CNN to segment tomograms and highlight voxels that belong to particles. After segmentation, a clustering step is applied to group activated voxels into individual particle coordinates. Other methods include cascades of U-Nets or incorporate denoising modules.

- **Weakly-Supervised and Positive-Unlabeled Learning.** To reduce the need of large labeled datasets, some methods use weak labels or small training sets and clever learning strategies. These approaches acknowledge that getting thousands of perfectly labeled particles in tomograms is infeasible, and instead try to learn from minimal, noisy or indirect annotations. For example, Topaz trains a 2D CNN with positive-unlabeled (PU) learning (use a few sparse positive labels manually picked and treats all other regions as "unlabeled" rather than definitely non-particle). Topaz demonstrated the ability to retrieve far more real particles from micrographs than traditional picking methods while maintaining low false positive rates. In cryo-ET, this method is not directly applicable out-of-the-box. A solution is to train a 3D CNN with handful of positive subvolumes and treating the rest of the tomogram as unlabeled background. This would alleviate the need for carefully labeling negatives. Indeed, annotation-efficient training is a theme in new cryo-ET pickers like DeepETPicker, which needs only a low number of "simplified labels".

2.5 Correlative Cryo-Electron Tomography (Cryo-CLEM)

The low signal-to-noise ratio and crowded nature of cryo-ET tomograms make global particle picking challenging. To address this, structural biologists often employ Correlative Cryo-Electron Tomography (Cryo-CLEM), a workflow that combines two or more

imaging modalities. Typically, this involves combining light microscopy (e.g., fluorescence microscopy, FM) with electron microscopy (EM).

In a common Cryo-CLEM workflow, light microscopy is first used on the cryogenically preserved specimen to identify regions of interest (ROIs) based on specific fluorescent markers (e.g., organelles or tagged protein locations). This low-resolution but high-contrast contextual information is then used to guide the high-resolution acquisition of the Cryo-ET tilt series.

While Cryo-SEM is rarely used for the final high-resolution tomogram itself, it is sometimes utilized as part of the overall correlative workflow, often in the form of **Focused Ion Beam (FIB)-milling**. This process uses the scanning capability of the SEM to image the sample's surface while the FIB ablates material to thin down a section of the cell (creating a lamella). The lamella is then thin enough for high-resolution TEM/Cryo-ET acquisition.

Therefore, the advantage of combining techniques is primarily **contextual** and **target-driven**: it allows researchers to dramatically reduce the search space, focusing particle picking efforts only on relevant cellular compartments that were pre-identified with the lower-resolution correlative methods.

2.6 Comparison: Cryo-TEM vs. Cryo-SEM

(Note: While both are cryogenic techniques, Cryo-ET relies primarily on TEM. Cryo-SEM is generally used for surface analysis, not 3D tomography of internal macromolecules.)

Transmission Cryo-Electron Microscopy (**Cryo-TEM**) analyzes the internal structure of the sample using high-energy electrons that transmit through the specimen. This is the standard modality for Single Particle Analysis and Tomography.

Scanning Cryo-Electron Microscopy (**Cryo-SEM**), in contrast, focuses on the sample surface using reflected or secondary electrons. While valuable for tissue morphology, it lacks the transmissive power to resolve internal atomic structures of individual protein complexes in the same way TEM does.

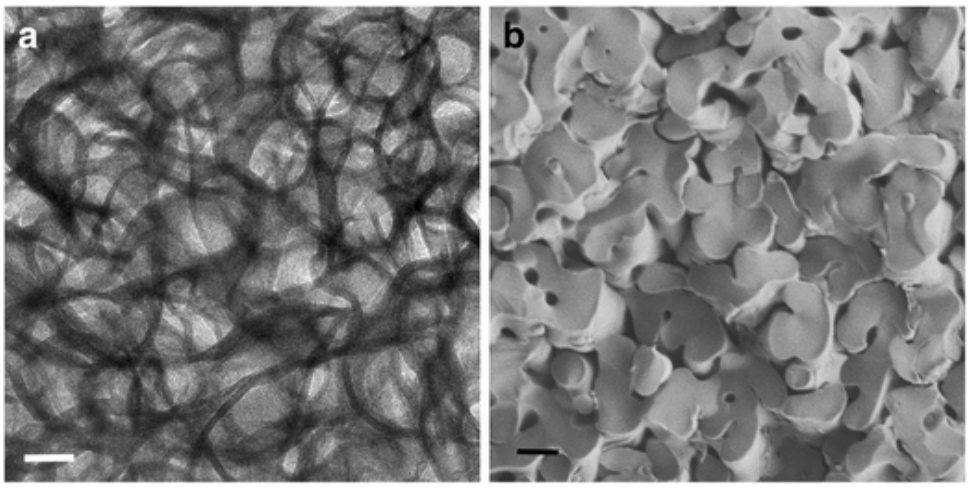

Figure 3: Cryo-TEM (a) and cryo-SEM (b) of the system at $\alpha = 80\,\%$ and $T = 36\,^{\circ}\mathrm{C}$, $7\,\mathrm{wt\%C_{12}E_5}$. Bars correspond to 500 nm.

3 Methodology

(Content to be added)

4 Results and Discussion

(Content to be added)

5 Conclusion and Future Work

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